

Kestrel Brakes Service Manual

Document: man-kestrel-brakes

Area: manuals

1 Introduction and Safety Precautions

This manual covers inspection and service of the Kestrel hydraulic brake system. Brake dust may contain hazardous material, so use approved cleaning methods and never blow dust with compressed air. Always support the vehicle on rated stands and verify a firm pedal before returning the vehicle to service.

Emissions-related cautions for shared underhood work appear in the Kestrel Emissions Service Manual , and the comparable electric-platform procedures are in the Marlin EV Brakes Service Manual .

2 Brake System Overview

The Kestrel braking system uses front pads BP-7720 on rotors BR-7731 fed by flexible hose BH-7745, with rear pads BP-7722 on rotors BR-7733. Hydraulic force originates at master cylinder BMC-7780 and is applied through front caliper BCAL-7760, using brake fluid BF-7790 throughout the circuit. The equivalent component layout for the electric platform is summarized in the Marlin EV Brakes Service Manual .

3 Front Brake Subsystem

The front brakes provide the majority of stopping force and consist of pads BP-7720, vented rotors BR-7731, and flexible hose BH-7745. Inspect the friction surface and hose routing at every brake service. Front-brake architecture on the electric platform is described in Marlin EV Brakes Service Manual :: Front Brakes .

3.1 Front Brake Pads and Rotors

Front pads BP-7720 should be replaced as an axle set whenever friction material falls below the minimum thickness. Rotors BR-7731 must be measured against the discard specification and resurfaced or replaced if below limit. Pad and rotor specifications for the comparable platform appear in Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications .

3.1.1 Front Brake Pad and Rotor Service Procedure

Procedure PROC-BRAKE-FR covers combined replacement of pads BP-7720 and rotors BR-7731. Remove the caliper and bracket, slide off the worn rotor, and clean the hub mating face before fitting the new BR-7731. Install pads BP-7720 with new hardware and bed them in per the burnishing schedule. The platform-equivalent combined service is documented in Marlin EV Brakes Service Manual :: Front Brake Pad and Rotor Service Procedure .

3.1.2 Front Brake Pad Replacement Procedure

Procedure PROC-BRAKE-PAD applies when only pads BP-7720 require replacement and the rotors remain within specification. Retract the caliper piston, remove the worn pads, and inspect the rotor surface for scoring before fitting new BP-7720 pads. Pump the pedal to seat the pads prior to road test. See Marlin EV Brakes Service Manual :: Front Brake Pad Replacement Procedure for the comparable steps.

3.2 Front Brake Hose

Flexible hose BH-7745 carries pressurized fluid from the chassis line to the moving caliper. Inspect BH-7745 for cracking, bulging, or chafing, and replace any hose showing damage. A compromised hose can cause uneven braking or fluid loss.

3.2.1 Brake Hose Replacement Procedure

Procedure PROC-BRAKE-HOSE directs the technician to clamp the supply line, disconnect both ends of hose

BH-7745, and cap the fittings to limit fluid loss. Install the new BH-7745 without twisting, torque the fittings to spec, and bleed the affected corner. Confirm there is no seepage under applied pedal pressure.

4 Rear Brake Subsystem

The rear brakes balance the vehicle under braking and consist of pads BP-7722 on rotors BR-7733. Inspect for even wear and proper caliper slide-pin function. Rear-brake layout for the electric platform is covered in Marlin EV Brakes Service Manual :: Rear Brakes .

4.1 Rear Brake Pads and Rotors

Rear pads BP-7722 and rotors BR-7733 should be inspected together and replaced as needed when either reaches its wear limit. Measure rotor BR-7733 thickness and lateral runout against specification. Comparable specifications are listed in Marlin EV Brakes Service Manual :: Rear Pad and Rotor Specifications .

4.1.1 Rear Brake Pad and Rotor Service Procedure

Procedure PROC-BRAKE-RR covers replacement of rear pads BP-7722 and rotors BR-7733. If the rear caliper integrates the parking brake, retract the piston with the correct wind-back tool rather than a straight push. Fit new rotor BR-7733 and pads BP-7722, then verify parking brake operation. The platform-equivalent procedure is Marlin EV Brakes Service Manual :: Rear Brake Pad and Rotor Service Procedure .

5 Brake Caliper Subsystem

The caliper clamps the pads against the rotor to convert hydraulic pressure into braking force. Front caliper BCAL-7760 is serviced as a unit when seized or leaking. Hydraulic actuation on the electric platform is described in Marlin EV Brakes Service Manual :: Hydraulic Actuation .

5.1 Front Brake Caliper

Front caliper BCAL-7760 should be inspected for piston seizing, boot damage, and fluid weep. A sticking BCAL-7760 causes uneven pad wear and pull under braking. Replace the caliper as an assembly when rebuild is not authorized.

5.1.1 Brake Caliper Replacement Procedure

Procedure PROC-CALIPER directs disconnecting the hydraulic line from caliper BCAL-7760, removing the mounting bolts, and transferring the bracket as required. Install the replacement BCAL-7760, torque the banjo fitting with new washers, and bleed the corner thoroughly. The comparable steps are in Marlin EV Brakes Service Manual :: Brake Caliper Replacement Procedure .

6 Hydraulic and Fluid Subsystem

The hydraulic subsystem generates and distributes brake line pressure using master cylinder BMC-7780 and brake fluid BF-7790. Maintain fluid at the correct level and replace on the specified interval to prevent moisture-induced fade.

6.1 Brake Master Cylinder

Master cylinder BMC-7780 converts pedal input into hydraulic pressure for both circuits. Inspect BMC-7780 for internal bypass, indicated by a slowly sinking pedal under steady pressure. Master cylinder service for the electric platform is covered in Marlin EV Brakes Service Manual :: Brake Master Cylinder Service .

6.1.1 Brake Fluid Flush and Bleed Procedure

Procedure PROC-BRAKE-FLUID specifies a complete flush of brake fluid BF-7790, bleeding each corner in the prescribed sequence until clean fluid flows free of air. Use only the specified BF-7790 grade and keep the reservoir topped throughout the process. Confirm a firm pedal and correct reservoir level before release.